Neural Machine Translation

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Abstract

This document outlines the implementation and evaluation of neural machine translation (NMT) models trained on the IWSLT 2013 French-to-English dataset. The assignment investigates the impact of subword tokenization methods, specifically Byte Pair Encoding (BPE), on translation quality. Using the Fairseq framework, two experiments are conducted: one employing BPE tokenization and the other using whole word tokenization via the Moses tokenizer. Results are evaluated using BLEU scores, with a focus on untokenized SacreBLEU to maintain standardization. The findings provide insights into tokenization's role in NMT performance and offer comparative analyses of translations with and without BPE.

Introduction

Neural machine translation (NMT) has become a cornerstone of natural language processing (NLP), leveraging deep learning to achieve state-of-the-art results in various language pairs. This assignment focuses on evaluating the influence of tokenization methods on the performance of NMT models, particularly for the IWSLT 2013 French-to-English dataset. Byte Pair Encoding (BPE) has been widely adopted in NMT for its ability to mitigate the out-of-vocabulary issue and reduce vocabulary size, while whole word tokenization, such as that provided by the Moses tokenizer, retains the full lexical representation of words.

The objective of this work is twofold: (1) train and evaluate a Transformer-based NMT model with BPE tokenization and analyze the resulting BLEU scores, and (2) retrain the model without BPE, using whole word tokenization, and compare the performance across key metrics. Through these experiments, the assignment aims to highlight the advantages and limitations of each tokenization strategy, offering practical insights into their impact on translation quality. The evaluation includes untokenized BLEU scores (SacreBLEU) for standardized assessment and a qualitative comparison of example translations from the development set.

Part 1: Training Models with Byte-Pair Encoding

In this section, two distinct models, a CNN-based and a Transformer-based encoder-decoder model, are trained using the Fairseq library with byte-pair encoding (BPE) applied during preprocessing. BPE helps reduce vocabulary size and mitigate out-of-vocabulary (OOV) issues by splitting words into subword units. BLEU scores (untokenized, using SacreBLEU) for the test split are reported, along with token counts for the train, validation, and test splits after BPE tokenization.

Results and Analysis

Table 1 presents the BLEU scores for both models, while Tables 2 and 3 provide token counts after BPE tokenization.

Table 1: BLEU Scores (untokenized, using SacreBLEU) for the fr-en Test Set with BPE

Model	BLEU Score
Transformer	28.19
CNN	26.88

Table 2: Unique Token Counts After Byte-Pair Encoding for Each Split in French (Source)

Split	Tokens
Train	8488
Validation	3450
Test	4014

Table 3: Unique Token Counts After Byte-Pair Encoding for Each Split in English (Target)

Split	Tokens
Train	7267
Validation	3128
Test	3442

Observations

- The Transformer model outperformed the CNN model on the test set, achieving a higher BLEU score of 28.19 compared to 26.88.
- Byte-pair encoding reduced the number of unique tokens, enabling the models to handle OOV words more effectively.
- The reduction in vocabulary size facilitated better optimization and faster convergence during training.

Part 2: Training Models Without Byte-Pair Encoding

In this section, the same CNN and Transformer models are retrained without using byte-pair encoding (BPE). Tokenization is performed directly on whole words using the Moses tokenizer. BLEU scores for the test set and token counts for the train, validation, and test splits without BPE are analyzed.

Results and Analysis

Table 4 presents the BLEU scores for both models, while Tables 5 and 6 provide token counts without BPE.

Table 4: BLEU Scores (untokenized, using SacreBLEU) for the fr-en Test Set Without BPE

Model	BLEU Score
Transformer	32.82
CNN	30.79

Table 5: Unique Token Counts Without Byte-Pair Encoding for Each Split in French (Source)

Split	Tokens
Train	64972
Validation	3731
Test	4633

Table 6: Unique Token Counts Without Byte-Pair Encoding for Each Split in English (Target)

Split	Tokens
Train	50804
Validation	3206
Test	3685

Observations

- Models trained without BPE achieved higher BLEU scores compared to Part 1, indicating the potential benefits of preserving full words during tokenization.
- The absence of BPE resulted in larger vocabulary sizes and token counts for each split.
- Example translations from the development set highlighted differences in translation quality, including cases where word-level tokenization captured better linguistic nuances compared to subword-based tokenization.

Development Set Translation Comparisons

Three example sentences from the development set showing better or worse translations in Part 2 compared to Part 1 are analyzed here.

- **Source (French):** *j' en porte presque tout le temps.*
- **Target Reference (English):** *I wear jeans almost all the time.*
- **Part 1 (With BPE):**
  - **Generated Translation:** *I 'm wearing it almost all the time.*
  - **Observation:** This translation introduces "it" instead of "jeans," which changes the meaning slightly but maintains overall fluency.
- **Part 2 (Without BPE):**
  - **Generated Translation:** *I 'm almost wearing it all the time.*
  - **Observation:** This translation misplaces "almost," altering the intended meaning and reducing alignment with the reference translation.
- **Source (French):** *quelle est cette chose ?*

- **Target Reference (English):** *And what is that thing?*
- **Part 1 (With BPE):**
  - **Generated Translation:** *What 's this thing?*
  - **Observation:** The translation is close to the reference but uses "this" instead of "that," leading to a slight shift in meaning while remaining natural.
- **Part 2 (Without BPE):**
  - **Generated Translation:** *What is that?*
  - **Observation:** This translation aligns with the reference in terms of meaning but omits "thing," resulting in a less complete sentence.
- **Source (French):** *il est absolument impossible de continuer sur cette tendance .*
- **Target Reference (English):** *This is a completely unsustainable pattern.*
- **Part 1 (With BPE):**
  - **Generated Translation:** *It 's absolutely impossible to keep on that trend.*
  - **Observation:** The translation is grammatically correct but deviates in meaning from the reference. Instead of describing the trend as "unsustainable," it translates to continuing on the trend, which alters the original intent.
- **Part 2 (Without BPE):**
  - **Generated Translation:** (Provide the No BPE result here if available, or leave this placeholder until added.)
  - **Observation:** (Comment on how the No BPE model's result differs or aligns with the target reference.)

Part 3

In this section, I explored modifications to the Transformer model and CNN models from Part 1 by experimenting with different numbers of layers and tuning the dropout rate. These adjustments were aimed at improving the model's performance, as measured by BLEU scores, on the French-to-English translation task.

Model Training Hyperparameters

The hyperparameters used to train the Transformer and CNN models in Part 1 are summarized in Tables 7 and 8, respectively. These values were chosen based on example configurations provided in the Fairseq README documentation, representing standard settings for training on the IWSLT dataset. Each table highlights the key parameters and settings used to achieve competitive results.

Table 7: Transformer Model Training Hyperparameters

Hyperparameter	Value
Architecture	transformer_iwslt_de_en
Embedding Sharing	Source and Target
Optimizer	Adam
Adam Betas	(0.9, 0.98)
Clip Norm	0.0
Learning Rate	5e-4
LR Scheduler	Inverse Square Root
Warmup Updates	4000
Dropout	0.3
Weight Decay	0.0001
Criterion	Label Smoothed Cross Entropy
Label Smoothing	0.1
Max Tokens	4096
BLEU Beam Size	5
Max Epochs	10

Table 8: CNN Model Training Hyperparameters

Hyperparameter	Value
Architecture	fconv_iwslt_de_en
Dropout Rate	0.1
Criterion	Label Smoothed Cross Entropy
Label Smoothing	0.1
Optimizer	NAG
Learning Rate	0.5
LR Scheduler	Fixed
Force Anneal	50
Gradient Clipping	0.1
Max Tokens	3000
Max Epochs	10
BLEU Beam Size	5
BPE Removal	Enabled

Hyperparameter Experiments for Transformers

In this section, I experiment with different hyperparameter settings for the number of encoder and

decoder layers and the dropout rate to observe their impact on the model's performance.

Experiment 1: Increased Layers with Moderate Dropout

- **Encoder Layers:** 6
- **Decoder Layers:** 6
- **Dropout Rate:** 0.05
- **Learning Rate:** 5e-4

Results:

- **BLEU Score:** 28.70
- **Observations:** The low dropout rate allowed the model to retain more information during training, which contributed to slightly improved performance. However, with 6 encoder and decoder layers, the improvement was not as significant as expected, possibly due to the complexity of the architecture not fully aligning with the dataset size or training configuration.

Experiment 2: Reduced Layers with Lower Dropout

- **Encoder Layers:** 5
- **Decoder Layers:** 5
- **Dropout Rate:** 0.2
- **Learning Rate:** 3e-4

Results:

- **BLEU Score:** 28.98
- **Observations:** Reducing the number of layers helped the model converge more effectively, particularly when combined with a slightly higher dropout rate. This configuration struck a balance between model capacity and generalization, resulting in the best BLEU score among the experiments.

Commentary

The experiments reveal that the dropout rate plays a significant role in balancing the model's ability to learn and generalize. Lower dropout rates (e.g., 0.05) allowed the model to retain more information, which was beneficial but introduced the risk of overfitting when combined with a higher number of layers. Conversely, slightly higher dropout rates (e.g., 0.2) improved the model's generalization, particularly when the number of encoder and decoder layers was reduced to 5.

Additionally, increasing the number of layers beyond 6 seemed to introduce diminishing returns and potentially overfitting, especially without a compensatory adjustment in dropout or other regularization techniques. On the other hand, reducing the number of layers made the model more efficient and, combined with a well-chosen dropout rate, led to the highest BLEU score (28.98). These findings highlight the importance of tuning hyperparameters carefully to achieve an optimal trade-off between learning capacity and generalization.

Hyperparameter Experiments for CNNs

In this section, I experiment with different hyperparameter settings for the number of encoder and decoder layers and the dropout rate to observe their impact on the CNN model's performance.

Experiment 1: Reduced Dropout with Custom Layer Configuration

- **Encoder Layers:** `[(512, 3)] * 4 + [(512, 4)] * 2`
- **Decoder Layers:** `[(512, 3)] * 4 + [(512, 4)] * 2`
- **Dropout Rate:** 0.05
- **Learning Rate:** 0.5

Results:

- **BLEU Score:** 27.55
- **Observations:** The reduced dropout rate helped retain more information during training, improving performance slightly compared to the baseline (26.88). The custom layer configuration allowed the model to focus more on localized patterns in the data, contributing to the modest improvement in BLEU score. However, additional tuning may be required to maximize the performance gains.

Experiment 2: Increased Layers with Higher Dropout

- **Encoder Layers:** `[(512, 3)] * 8 + [(512, 4)] * 2`
- **Decoder Layers:** `[(512, 3)] * 8 + [(512, 4)] * 2`

- **Dropout Rate:** 0.2
- **Learning Rate:** 0.5

Results:

- **BLEU Score:** 27.67
- **Observations:** Increasing the number of encoder and decoder layers to 10 while raising the dropout rate to 0.2 resulted in a slight improvement in BLEU score. The deeper architecture seems to have captured more complex patterns, but the performance gain was not significant compared to Experiment 1. The higher dropout might have helped mitigate overfitting but could also have limited the model's ability to fully utilize the added capacity.

Commentary

The CNN experiments demonstrate that hyperparameter tuning, such as adjusting dropout rates and layer configurations, can have a noticeable impact on performance. Experiment 1 revealed that lowering the dropout rate to 0.05 improved performance slightly, suggesting that the model benefited from retaining more information during training. In contrast, Experiment 2, which involved a deeper architecture with a higher dropout rate of 0.2, showed a marginal increase in BLEU score, likely due to the balance between added capacity and regularization.

Future experiments could focus on further adjustments to the layer configurations, learning rate, or dropout rate to optimize the model's performance. Additionally, combining these experiments with other techniques, such as tuning kernel sizes or increasing batch sizes, could yield more significant improvements.

Overall

The best performance achieved during the hyperparameter optimization experiments was a BLEU score of **28.98** (untokenized, using SacreBLEU) for the Transformer model. This result was obtained by reducing the encoder and decoder layers to **5 each** and setting the dropout rate to **0.2**. The learning rate for this configuration was set to **3e-4**.

For the CNN model, the highest BLEU score achieved was **27.67**, using a deeper layer configuration of `[(512, 3)] * 8 + [(512, 4)] * 2` for both the encoder and decoder, with a dropout rate of **0.2** and learning rate of **0.5**.

These experiments demonstrate that tuning dropout rates and adjusting the number of encoder and decoder layers significantly impacts the performance of neural machine translation models. The findings highlight the importance of balancing model capacity and regularization to optimize results.

Development Set Translation Comparisons

This section compares translations generated by the best Transformer model from Part 3 to the baseline Transformer model from Part 1. Three examples where the Part 3 model provided better translations and one example where it performed worse are presented. Each comparison includes the source sentence, reference translation, and the outputs from both models, along with observations.

Example 1: Better Translation

Source (French): *vous avez peut-être lu que nous avons mis des filigranes.*
Reference (English): *You may have read that we put watermarks in.*
Part 1 (Baseline): *You may read it, we insisted signs.*
Part 3 (Best Model): *Maybe you read it, we signed signatures.*
Observation: The Part 3 translation is more coherent and contextually accurate than the baseline. While the baseline ("we insisted signs") introduces grammatical errors and loses the intended meaning, the Part 3 model maintains better fluency and conveys a clearer message, despite slight deviations from the reference translation.

Example 2: Better Translation

Source (French): *par conséquent, le téléphone n' était jamais cassé.*
Reference (English): *One consequence of that, by the way, is that the phone never broke.*
Part 1 (Baseline): *So the phone was never broken.*
Part 3 (Best Model): *As a result, the phone was never broken.*
Observation: The Part 3 translation improves upon the baseline by using the phrase "As a result," which better captures the causal relationship implied in the source sentence. This aligns more closely with the reference, both semantically and stylistically, compared to the simpler and less contextually appropriate "So" used in the baseline.

Example 3: Better Translation

Source (French): *superposées à une image géospatiale de la nasa.*
Reference (English): *Superimposed on top of a NASA geospatial image.*
Part 1 (Baseline): *Suppose a geospace image of the NASA.*
Part 3 (Best Model): *Superposed to a geospatial image of the NASA.*
Observation: The Part 3 translation improves upon the baseline by producing a grammatically correct and semantically accurate translation. While the phrase "superposed to" is slightly less natural than "superimposed on," it closely matches the meaning of the reference translation. The baseline translation, in contrast, introduces an unrelated and incorrect word, "Suppose," which significantly alters the meaning of the sentence.

Example 4: Worse Translation

Source (French): *frosty le charbon est un joyeux bonhomme.*
Reference (English): *Frosty the coal man is a jolly, happy soul.*
Part 1 (Baseline): *Crusty coal is a nice man.*
Part 3 (Best Model): *Africany coal is a joke.*
Observation: The Part 3 translation introduces a significant error, replacing "Frosty" with "Africany" and "jolly, happy soul" with "a joke," which drastically alters the meaning and tone of the sentence. In contrast, the baseline translation, while not perfect, maintains a closer semantic alignment to the reference by describing "Crusty coal" as "a nice man."

Summary of Observations

The examples above demonstrate the improvements achieved by the best Transformer model from Part 3 over the baseline model from Part 1. The Part 3 model excelled in areas such as [highlight key improvements, e.g., fluency, contextual understanding, or alignment with reference translations], while occasionally underperforming in cases like [highlight key challenges, e.g., rare words or complex structures]. These results underscore the impact of hyperparameter tuning on translation quality.